



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2004

**TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- All *necessary* working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Hand in your answer booklets in 3 sections.
Section A (Questions 1 - 3),
Section B (Questions 4 - 5) and
Section C (Questions 6 - 8).
- Start each section in a **NEW** answer booklet.

Total Marks - 120 Marks

- Attempt Sections A - C
- All questions are NOT of equal value.

Examiner: *E. Choy*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Total marks – 120
Attempt Questions 1 – 8
All questions are of equal value

Answer each section in a SEPARATE writing booklet. Extra writing booklets are available.

SECTION A (Use a SEPARATE writing booklet)

Question 1 (15 marks)		Marks
(a)	Evaluate	
(i)	$\int_0^{\sqrt{3}} \frac{dx}{\sqrt{4-x^2}}$	1
(ii)	$\int_0^1 \sqrt{4-x^2} dx$	1
(iii)	$\int_{-1}^2 x\sqrt{2-x} dx$	1
(b)	Evaluate	
(i)	$\int_1^2 \frac{e^{2x}}{e^x-1} dx$	2
(ii)	$\int_0^{\frac{\pi}{2}} \frac{1}{4+5\sin x} dx$	4
(c)	(i) If $I_n = \int_0^{\frac{\pi}{4}} \tan^{2n} x dx$, $n \geq 0$, show that $I_n + I_{n-1} = \frac{1}{2n-1}$	3
	(ii) Hence, evaluate $\int_0^{\frac{\pi}{4}} \tan^6 x dx$	1
(d)	Evaluate $\int_1^e x \ln(x^2) dx$	2

Question 2 (15 marks)

Marks

- (a) (i) Sketch on the same axes the graphs

2

$$y = x + 3 \text{ and } y = 2|x|.$$

- (ii) Hence or otherwise:

(α) Solve for x , $2|x| < x + 3$.

2

(β) Sketch the curve $y = \frac{2|x|}{x+3}$.

3

(b) Let $f(x) = \frac{3}{x-1}$.

On separate diagrams sketch the graphs of the following:

(i) $y = f(|x|)$

2

(ii) $y^2 = f(x)$

3

(iii) $y = e^{f(x)}$

3

SECTION A continued

Question 3 (15 marks)

Marks

- (a) If $z = -1 + i\sqrt{3}$ and $w = 2\text{cis}\frac{\pi}{6}$
- (i) Find $|z|$. 1
- (ii) $\arg z$. 1
- (iii) Express z in the form $r\text{cis}\theta$. 1
- (iv) Express $z^6 \div w^3$ in the form $r\text{cis}\theta$. 1
-
- (b) (i) Express $\sqrt{5-12i}$ in the form $a+ib$. 2
- (ii) Hence describe the locus of the point which represents z on the Argand diagram if 2
- $$|z^2 - 5 + 12i| = |z - 3 + 2i|$$
-
- (c) The origin and the points representing the complex numbers z , $\frac{1}{z}$ and $z + \frac{1}{z}$ are joined to form a quadrilateral. Write down the conditions for z so that the quadrilateral will be
- (i) a rhombus; 1
- (ii) a square. 1
-
- (d) (i) Find the equation and sketch the locus of z if 2
- $$|z - i| = \text{Im}(z)$$
- (ii) Find the least value of $\arg z$ in (i) above. 3

END OF SECTION A

SECTION B (Use a SEPARATE writing booklet)

Question 4 (15 marks)

Marks

- (a) $3-i$ is a zero of $P(z) = z^3 - 4z^2 - 2z + m$, where m is a real number.

3

Find m .

- (b) If α , β and γ are the roots of $x^3 + px + q = 0$, find a cubic equation whose roots are α^2 , β^2 and γ^2 .

3

- (c) Given a real polynomial $Q(x)$, show that if α is a root of $Q(x) - x = 0$, then α is also a root of $Q(Q(x)) - x = 0$.

3

- (d) Use the following identity to answer the following questions.

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$$

- (i) Solve $16x^5 - 20x^3 + 5x = 0$

3

- (ii) Hence show that

3

$$\cos \frac{\pi}{10} \cos \frac{3\pi}{10} \cos \frac{7\pi}{10} \cos \frac{9\pi}{10} = \frac{5}{16}$$

SECTION B continued

Question 5 (15 marks)

Marks

(a) Let $z = \cos \theta + i \sin \theta$, show that

(i) $z^n + z^{-n} = 2 \cos n\theta$ 1

(ii) $z^n - z^{-n} = 2i \sin n\theta$ 1

(b) (i) Show that for any integer k that 2

$$\left[z - \left(\cos \frac{k\pi}{4} + i \sin \frac{k\pi}{4} \right) \right] \left[z - \left(\cos \frac{(8-k)\pi}{4} + i \sin \frac{(8-k)\pi}{4} \right) \right] = z^2.$$

(ii) Hence simplify the following products

(α) $\left[z - \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \right] \left[z - \left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} \right) \right]$ 1

(β) $\left[z - \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right) \right] \left[z - \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right]$ 1

(c) Using the results of (b) above, factorise $z^4 + 1$ into 2 real quadratic factors. 2

(d) Using (a) and (c) above, prove the identity 2

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

(e) The complex numbers $z = x + iy$, $z_1 = -x + iy$ and $z_2 = -\frac{2}{z}$ are represented by the points P , P_1 and P_2 in the Argand diagram respectively.

(i) Show that O , P_1 and P_2 are collinear where O is the origin. 3

(ii) Show that $OP_1 \times OP_2 = 2$ 2

END OF SECTION B

SECTION C (Use a SEPARATE writing booklet)

Question 6 (15 marks)

Marks

- (a) A particle of mass m is projected vertically upwards with a velocity of $u \text{ ms}^{-1}$, with air resistance proportional to its velocity.

- (i) Show that after a time t seconds, the height above the ground is

4

$$x_1 = \frac{g + ku}{k^2} (1 - e^{-kt}) - \frac{gt}{k},$$

where k is a constant and g is the acceleration due to gravity.

- (ii) At the same time another particle of mass m is released from rest, from a height h metres vertically above the first particle. You may assume that at time t seconds, its distance from the ground is given by:

4

$$x_2 = h + \frac{g}{k^2} (1 - e^{-kt}) - \frac{gt}{k}$$

Show that the two particles will meet at time T where

$$T = \frac{1}{k} \ln \left(\frac{u}{u - kh} \right)$$

- (b) A vehicle of mass m moves in a straight line subject to a resistance $P + Qv^2$, where v is the speed and P and Q are constants with $Q > 0$.

- (i) Form an equation of motion for the acceleration of the vehicle.

1

- (ii) Hence show that if $P = 0$, the distance required to slow down from speed $\frac{3U}{2}$ to speed U is $\frac{m}{Q} \ln \left(\frac{3}{2} \right)$.

3

- (iii) Also show that if $P > 0$, the distance required to stop from speed U is given by

3

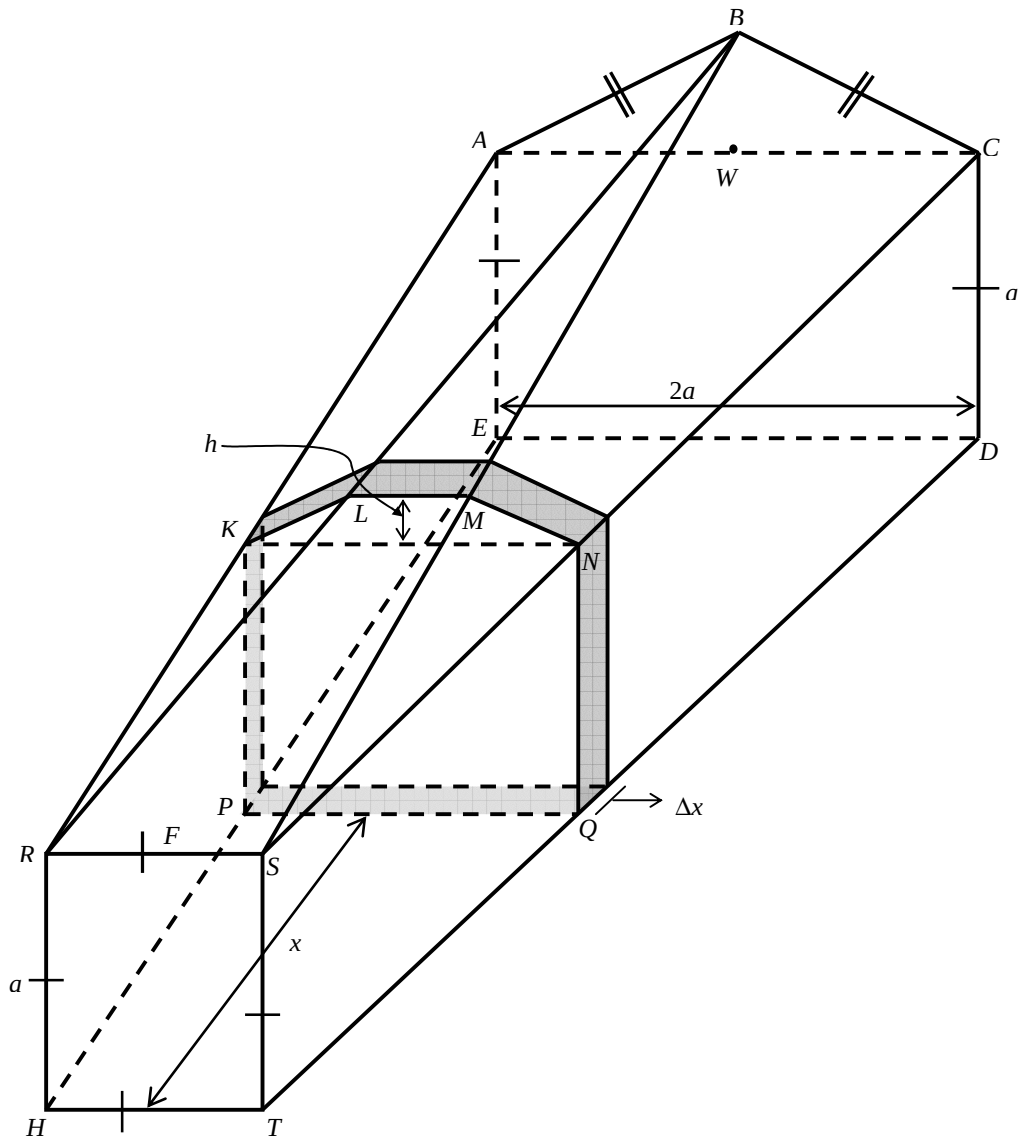
$$D = \lambda \ln(1 + kU^2)$$

where k and λ are constants

SECTION C continued

Question 7 (15 marks)

Marks



The diagram above shows a solid with a trapezoidal base $EDTH$ of length b metres.

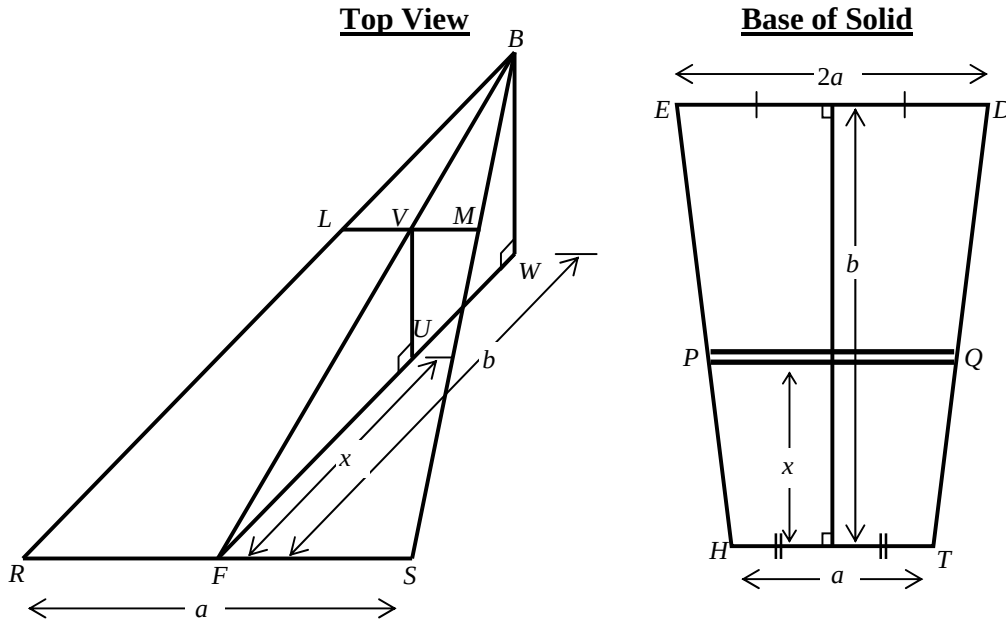
The front end $HTSR$ is a square with side length a metres.

The back is the pentagon $ABCDE$ which consists of the rectangle $ACDE$ with length $2a$ metres and width a metres, surmounted by the equilateral triangle ABC .

Consider a slice of the solid, parallel to the front and the back, with face formed by both the trapezium $KLMN$ and the rectangle $KNQP$, which has thickness Δx and is at a distance x metres from HT .

Question 7 continued on page 9

- (i) Show that the height, BW , of the equilateral triangle ABC is $\sqrt{3}a$ metres. 2



- (ii) Given that the perpendicular height of the trapezium $KLMN$ is h metres ie $VU = h$, use the similar triangles BWF and VUF , in the Top View, to find h in terms of a , b and x . 3
- (iii) Given that the triangles BLM and BRS are similar, show that 3
- $$LM = \frac{a(b-x)}{b}$$
- (iv) Using the cross section of the base, find the length of PQ in terms of a , b and x . 3
- (v) Find the volume of the solid. 4

Question 8 starts on page 10

SECTION C continued

Question 8 (15 marks)

Marks

- | | | |
|-------|--|---|
| (a) | If $a > 0$, $b > 0$ and $a + b = t$ show that
$\frac{1}{a} + \frac{1}{b} \geq \frac{4}{t}$ | 3 |
| | | |
| (b) | There are n ($n > 1$) different boxes each of which can hold up to $n + 2$ books. Find the probability that: | |
| (i) | No box is empty when n different books are put into the boxes at random. | 1 |
| (ii) | Exactly one box is empty when n different books are put into the boxes at random. | 2 |
| (iii) | No box is empty when $n + 1$ different books are put into the boxes at random. | 2 |
| (iv) | No box is empty when $n + 2$ different books are put into the boxes at random. | 2 |
| | | |
| (c) | <p>$PQRS$ is a cyclic quadrilateral such that the sides PQ, QR, RS and SP touch a circle at A, B, C and D respectively.</p> <p>Prove that:</p> | |
| (i) | AC is perpendicular to BD . | 2 |
| (ii) | <p>Let the midpoints of AB, BC, CD and DA be E, F, G and H respectively.</p> <p>Show that E, F, G and H lie on a circle.</p> | 3 |

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$



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Sample Solutions

Section	Marker
A	Mr Hespe
B	Mr Kourtesis
C	Mr Parker

Section A

$$1(a)(i) \ I = \left[\sin^{-1} \frac{z}{2} \right]_0^{\sqrt{3}},$$

$$= \pi/3.$$

$$(ii) \ I = \int_0^{\pi/6} 4 \cos^2 \theta \, d\theta, \quad \text{put } x = 2 \sin \theta \quad x=1, \theta=\pi/6$$

$$dx = 2 \cos \theta \, d\theta \quad x=0, \theta=0$$

$$= 2 \int_0^{\pi/6} (1 + \cos 2\theta) \, d\theta,$$

$$= 2 \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\pi/6},$$

$$= \frac{\pi}{3} + \frac{\sqrt{3}}{2}.$$

$$(iii) \ I = \int_0^3 (2-u) u^{1/2} \, du, \quad \text{put } u = 2-x \quad x=-1, u=3$$

$$du = -dx \quad x=2, u=0$$

$$= \int_0^3 (2u^{1/2} - u^{3/2}) \, du,$$

$$= \left[\frac{4u^{3/2}}{3} - \frac{2u^{5/2}}{5} \right]_0^3,$$

$$= 4\sqrt{3} - \frac{18\sqrt{3}}{5},$$

$$= \frac{2\sqrt{3}}{5}.$$

$$\begin{aligned}
 1 \text{ (b)(i)} \quad I &= \int_{e^{-1}}^{e^2-1} \frac{1+u}{u} du, \quad \text{put } u = e^x - 1 \quad x=1, u=e-1 \\
 &\quad \quad \quad du = e^x dx \quad x=2, u=e^2-1 \\
 &= \left[\ln u + u \right]_{e^{-1}}^{e^2-1}, \\
 &= \ln(e^2-1) + e^2 - 1 - \ln(e-1) (e-1), \\
 &= e^2 - e + \ln(e+1).
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad I &= \int_0^1 \frac{1}{4+5x+\frac{2x}{1+x^2}} \times \frac{2dt}{1+t^2}, \quad \text{put } t = \tan \frac{x}{2} \\
 &\quad \quad \quad dt = \frac{1}{2} \sec^2 \frac{x}{2} dx \\
 &= \int_0^1 \frac{dt}{2+2t^2+5t}, \quad \text{ie } dx = \frac{2dt}{1+t^2} \\
 &= \int_0^1 \frac{dt}{(2t+1)(t+2)}. \quad \begin{matrix} x=0, t=0 \\ x=\frac{\pi}{2}, t=1 \end{matrix}
 \end{aligned}$$

$$\begin{aligned}
 \text{Now, } \frac{1}{(2t+1)(t+2)} &= \frac{A}{2t+1} + \frac{B}{t+2} \\
 1 &= A(t+2) + B(2t+1) \\
 \text{Let } t &= -\frac{1}{2}, \quad 3A/2 = 1, \quad A = 2/3 \\
 t &= -2, \quad -3B = 1, \quad B = -1/3
 \end{aligned}$$

$$\begin{aligned}
 \text{So, } I &= \frac{1}{3} \int_0^1 \left(\frac{2}{2t+1} - \frac{1}{t+2} \right) dt, \\
 &= \frac{1}{3} \left[\ln(2t+1) - \ln(t+2) \right]_0^1, \\
 &= \frac{1}{3} \left\{ \ln \frac{3}{1} - \ln \frac{3}{2} \right\}, \\
 &= \frac{\ln 2}{3}.
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)(i)} \quad I_n &= \int_0^{\pi/4} (\tan^2 x)^n dx, \\
 &= \int_0^{\pi/4} (\tan^2 x)^{n-1} (\sec^2 x - 1) dx, \\
 &= \int_0^{\pi/4} \tan^{2n-2} x d \tan x - I_{n-1}, \\
 I_n + I_{n-1} &= \left[\frac{\tan^{2n-1} x}{2n-1} \right]_0^{\pi/4}, \\
 &= \frac{1}{2n-1}.
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad I_3 &= \frac{1}{5} - I_2, \\
 I_2 &= \frac{1}{3} - I_1 = \frac{1}{3} - \int_0^{\pi/4} \tan^2 x dx \\
 &= \frac{1}{3} - \int_0^{\pi/4} (\sec^2 x - 1) dx
 \end{aligned}$$

$$I_2 = \frac{1}{3} - \left[\tan x - x \right]_0^{\pi/4},$$

$$= \frac{1}{3} - 1 + \frac{\pi}{4}.$$

$$\therefore I_3 = \frac{1}{5} + \frac{2}{3} - \frac{\pi}{4},$$

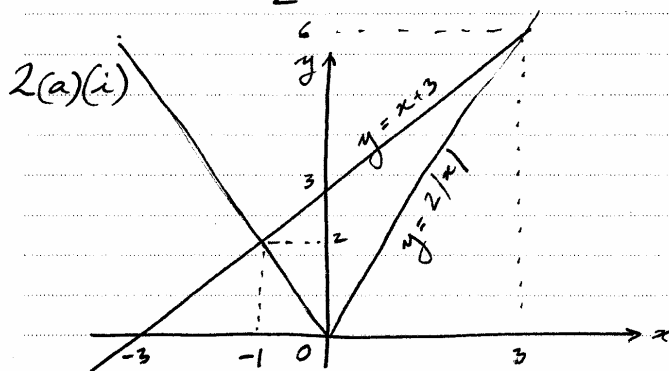
$$= \frac{13}{15} - \frac{\pi}{4}.$$

$$1(a) I = \left[x^2 \ln x \right]_1^e - \int_1^e x dx, \quad \begin{array}{l} u = \ln(x^2) \quad v' = x \\ u' = \frac{2}{x} \quad v = \frac{x^2}{2} \end{array}$$

$$= e^2 - 0 - \left[\frac{x^2}{2} \right]_1^e,$$

$$= e^2 - \frac{e^2}{2} + \frac{1}{2},$$

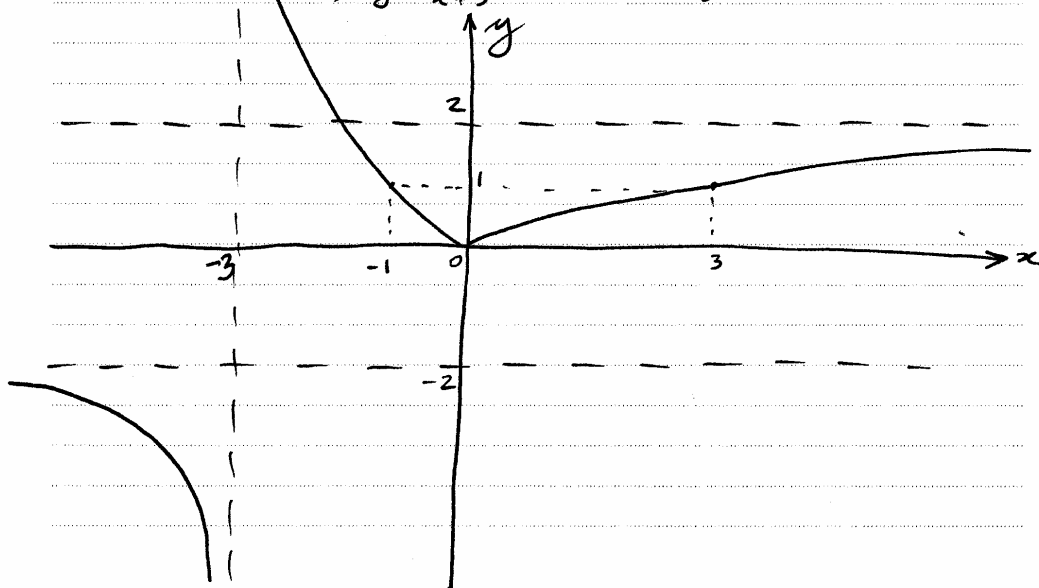
$$= \frac{e^2 + 1}{2}.$$



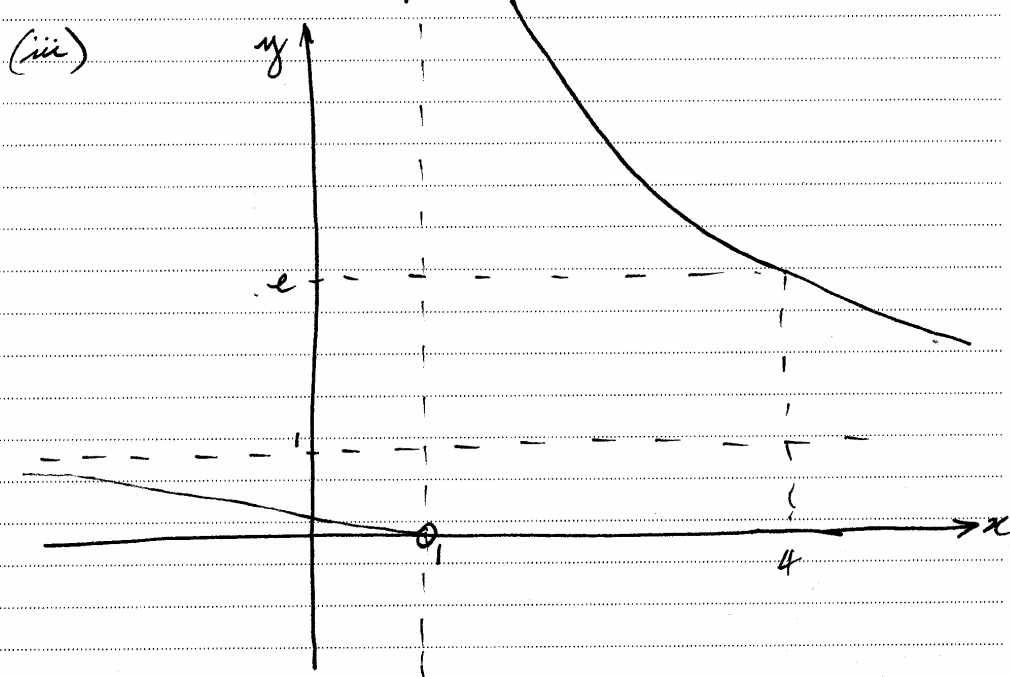
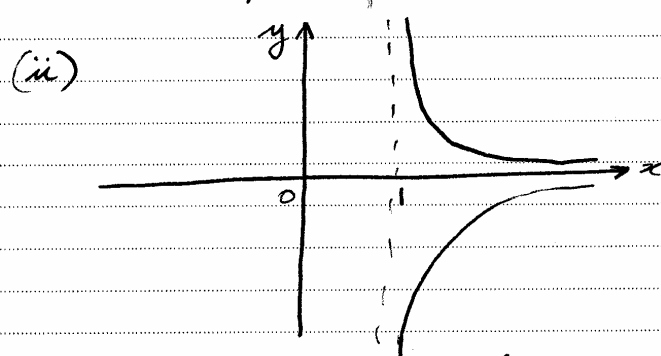
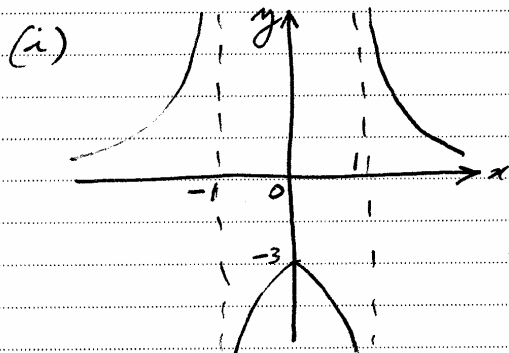
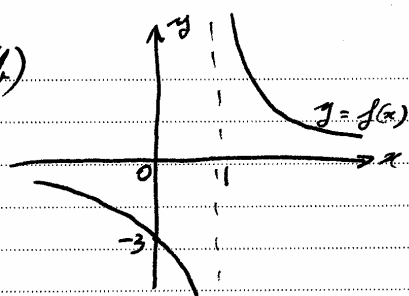
(ii) (a) From the graph $-1 < x < 3$.

(b) If $x < 0$, $y = \frac{-2x}{x+3} = -2 + \frac{6}{x+3}$,

$x > 0$, $y = \frac{2x}{x+3} = 2 - \frac{6}{x+3}$.



2(b)



$$3(a)(i) |z| = \sqrt{1+3},$$

$$= 2.$$

$$(ii) \arg z = \tan^{-1}\left(\frac{\sqrt{3}}{-1}\right),$$

$$= \frac{2\pi}{3}.$$

$$(iii) z = 2 \operatorname{cis} 2\pi/3.$$

$$(iv) z^6 \div w^3 = \frac{2^6}{2^3} \operatorname{cis}\left(6 \times \frac{2\pi}{3} - 3 \times \frac{\pi}{6}\right)$$

$$= 8 \operatorname{cis}(7\pi/2) \text{ or } 8 \operatorname{cis}(-\pi/2)$$

$$= -8i.$$

$$(b)(i) 5-12i = a^2+2abi-b^2$$

$$a^2+b^2 = 13$$

$$a^2-b^2 = 5$$

$$ab = -6$$

$$2a^2 = 18$$

$$a = \pm 3$$

$$b = \mp 2$$

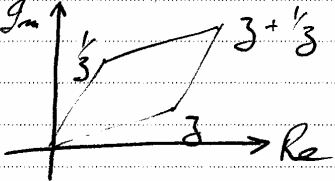
$$\therefore \sqrt{5-12i} = \pm(3-2i)$$

$$3(b)(ii) \quad z^2 - (5-12i) = z^2 - (3-2i)^2$$

$$= (z+3-2i)(z-3+2i)$$

$$\frac{|z+3-2i|}{|z-3+2i|} = \frac{|z-3+2i|}{|z+3-2i|} = 1$$

So locus of z is a circle centre $(-3, 2)$, radius 1.

(c)(i)  $|z| = |\frac{1}{z}|$ for a rhombus

so if $z = r \operatorname{cis} \theta$
 $r = \frac{1}{r}$
 $r^2 = 1$
 $r = 1$ (taking +ve root)

$\therefore |z| = 1$ is the condition.

(ii) $\pm iz = \frac{1}{z}$

$$\theta \pm \frac{\pi}{2} = -\theta + n\pi$$

$$2\theta = \pm \frac{\pi}{2} + n\pi$$

$$\theta = \pm \frac{\pi}{4}, \pm \frac{3\pi}{4}$$

$\therefore$ Conditions are $|z| = 1$

$$\arg z = \pm \frac{\pi}{4}, \pm \frac{3\pi}{4}$$

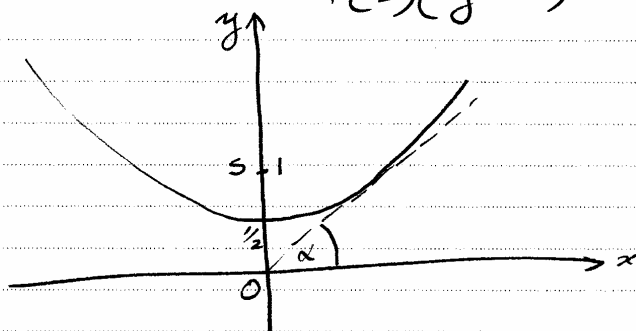
(d)(i) Let $z = x + iy$

$$\sqrt{x^2 + (y-1)^2} = y$$

$$x^2 + y^2 - 2y + 1 = y^2$$

$$x^2 = 2y - 1$$

$$= 4\left(\frac{1}{2}\right)\left(y - \frac{1}{2}\right)$$



(ii) $y = \frac{x^2+1}{2}$, parabola

$y = mx$, tangent
 $mx = \frac{x^2+1}{2}$

$$x^2 - 2mx + 1 = 0$$

$\Delta = 0$ at tangent

$$4m^2 - 4 = 0$$

$$m = \pm 1$$

$$\alpha = \pi/4, 3\pi/4$$

$\therefore$ Minimum argument $\pi/4$.

Section B

Question 4

$$\begin{aligned}
 (a) \quad P(3-i) &= (3-i)^3 - 4(3-i)^2 - 2(3-i) + m = 0 \\
 &= 18 - 26i - 32 + 24i - 6 + 2i + m = 0 \\
 &\Rightarrow (m-20) + i(0) = 0 \\
 \therefore m &= 20
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad \text{Required equation is } P(\sqrt{x}) &= (\sqrt{x})^3 + p\sqrt{x} + q = 0 \\
 \text{i.e. } x\sqrt{x} + p\sqrt{x} &= -q \\
 \Rightarrow [x\sqrt{x} + p\sqrt{x}]^2 &= [-q]^2 \\
 \therefore x^3 + 2px^2 + p^2x - q^2 &= 0
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad \text{If } \alpha \text{ is a root } &\Rightarrow Q(\alpha) - \alpha = 0 \\
 \text{i.e. } Q(\alpha) &= \alpha \\
 \text{Now } Q[Q(\alpha)] - \alpha &= Q[\alpha] - \alpha \\
 &= \alpha - \alpha \\
 &= 0
 \end{aligned}
 \left. \vphantom{\begin{aligned} Q[Q(\alpha)] - \alpha &= Q[\alpha] - \alpha \\ &= \alpha - \alpha \\ &= 0 \end{aligned}} \right\} \text{ using } Q(\alpha) = \alpha$$

$$(d) (i) \text{ let } x = \cos \theta \Rightarrow \cos 5\theta = 16x^5 - 20x^3 + 5x = 0$$

$$\text{Now } \cos 5\theta = 0$$

$$\Rightarrow 5\theta = 2k\pi \pm \frac{\pi}{2}$$

$$\theta = \frac{2k\pi}{5} \pm \frac{\pi}{10}$$

$$\text{using } \theta = \frac{2k\pi}{5} + \frac{\pi}{10} \quad \text{when } k=0, \theta = \frac{\pi}{10}$$

$$k=1, \theta = \frac{\pi}{2}$$

$$k=-1, \theta = -\frac{3\pi}{10}$$

$$k=2, \theta = \frac{9\pi}{10}$$

$$k=-2, \theta = -\frac{7\pi}{10}$$

$$\text{Now roots of } 16x^5 - 20x^3 + 5x = 0$$

$$\text{are of the form } x = \cos \theta \text{ i.e. } x = \cos \frac{\pi}{10}, \cos \frac{\pi}{2}, \cos(-\frac{3\pi}{10}), \cos \frac{9\pi}{10}, \cos(-\frac{7\pi}{10})$$

$$\Rightarrow \text{Roots are } \cos \frac{\pi}{10}, \cos \frac{3\pi}{10}, \cos \frac{9\pi}{10}, \cos \frac{7\pi}{10}, \cos \frac{\pi}{2}$$

$$\left[\text{Since } \cos \theta = \cos(-\theta) \right]$$

$$\begin{aligned}
 (ii) \quad 16x^4 - 20x^2 + 5 &= 0 \text{ has roots} \\
 x = \cos \frac{\pi}{10}, \cos \frac{3\pi}{10}, \cos \frac{7\pi}{10}, \cos \frac{9\pi}{10} &\therefore \cos \frac{\pi}{10} \cdot \cos \frac{3\pi}{10} \cdot \cos \frac{7\pi}{10} \cdot \cos \frac{9\pi}{10} = \frac{5}{16} \quad \text{Prod. of Roots}
 \end{aligned}$$

Question 5

$$(a) z^n = \cos(n\theta) + i\sin(n\theta) \quad \text{--- (A)}$$

$$z^{-n} = \cos(-n\theta) + i\sin(-n\theta)$$

$$\therefore z^{-n} = \cos(n\theta) - i\sin(n\theta) \quad \text{--- (B)}$$

$$(i) z^n + z^{-n} = 2\cos n\theta \quad \text{(A)+(B)}$$

$$(ii) z^n - z^{-n} = 2i\sin n\theta \quad \text{(A)-(B)}$$

(b)(i)

$$\begin{aligned} & \left[z^2 - z i s \frac{k\pi}{4} \right] \left[z - i s \left(\frac{8-k}{4} \right) \pi \right] \\ &= z^2 - \left[i s \frac{k\pi}{4} + i s \left(\frac{8-k}{4} \right) \right] z + \left[i s \frac{k\pi}{4} \cdot i s \left(\frac{8-k}{4} \right) \right] \\ &= z^2 - 2 \cos \frac{k\pi}{4} z + i s 0 \quad \text{(from (a))} \\ &= z^2 - 2 \cos \frac{k\pi}{4} z + 1 \end{aligned}$$

(ii) let $k=1$ in (i)

$$\begin{aligned} \Rightarrow \text{LHS} &= z^2 - 2z \cos \frac{\pi}{4} + 1 \\ &= z^2 - \frac{2z}{\sqrt{2}} + 1 \end{aligned}$$

(ii) let $k=3$ in (i)

$$\begin{aligned} \Rightarrow \text{LHS} &= z^2 - 2z \cos \frac{3\pi}{4} + 1 \\ &= z^2 + \frac{2z}{\sqrt{2}} + 1 \end{aligned}$$

(c) Since non-real zeros occur in conjugate pairs $\Rightarrow$ quadratic factors

$$\begin{aligned} z^4 + 1 &= [z - i s \frac{\pi}{4}] [z - i s \frac{3\pi}{4}] [z - i s \frac{5\pi}{4}] [z - i s \frac{7\pi}{4}] \\ &= \left[z^2 - \frac{2z}{\sqrt{2}} + 1 \right] \left[z^2 + \frac{2z}{\sqrt{2}} + 1 \right] \end{aligned}$$

using (ii)

$$= \left[z^2 - \frac{2z}{\sqrt{2}} + 1 \right] \left[z^2 + \frac{2z}{\sqrt{2}} + 1 \right]$$

$$(d) z^2 + z^{-2} = 2\cos 2\theta$$

$$\frac{z^4 + 1}{z^2} = 2\cos 2\theta$$

$$z^4 + 1 = 2z^2 \cos 2\theta$$

$$[z + \frac{1}{z}] [z - \frac{1}{z}] = 2\cos 2\theta$$

$$\left[\left(z + \frac{1}{z} \right) + \sqrt{2} \right] \left[\left(z + \frac{1}{z} \right) - \sqrt{2} \right] = 2\cos 2\theta$$

$$[2\cos \theta + \sqrt{2}] [2\cos \theta - \sqrt{2}] = 2\cos 2\theta$$

$$4\cos^2 \theta - 2 = 2\cos 2\theta$$

$$2\cos^2 \theta - 1 = \cos 2\theta$$

(e)

$$(i) \text{ let } \arg z = \theta$$

$$\therefore \arg z_1 = \pi - \theta$$

$$\arg z_2 = \arg \left(-\frac{2}{z} \right) = \arg(-2) - \arg(z)$$

$$\Rightarrow \arg z_2 = \pi - \theta$$

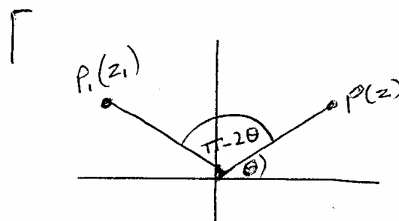
$$\arg z_1 = \arg z_2 = \pi - \theta$$

Since $|z_1| \neq |z_2|$ then O, P_1, P_2 collinear
 $[z_1 = |z_1| \arg(\pi - \theta); z_2 = |z_2| \arg(\pi - \theta)]$

$$(ii) OP_1 = \sqrt{(-x)^2 + (y)^2} = \sqrt{x^2 + y^2}$$

$$OP_2 = \sqrt{\left(\frac{-2x}{x^2 + y^2} \right)^2 + \left(\frac{2y}{x^2 + y^2} \right)^2} = \frac{2}{\sqrt{x^2 + y^2}}$$

$$\Rightarrow OP_1 \times OP_2 = \sqrt{x^2 + y^2} \times \frac{2}{\sqrt{x^2 + y^2}} = 2$$



$$\arg z = \theta$$

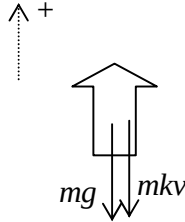
$$\arg(z_1 - 0) = \pi - \theta$$

$$\arg(z_2 - 0) = \pi - \theta$$

Section C

Q6

- (a) With $R \propto v$, to make the algebra easier take $R = mkv$



$$\begin{aligned}
 \text{(i)} \quad m \frac{dv}{dt} &= -(mg + mkv) \\
 \frac{dv}{dt} &= -(g + kv) \Rightarrow \frac{dt}{dv} = -\frac{1}{g + kv} = -\frac{1}{k} \left(\frac{k}{g + kv} \right) \\
 \therefore t &= -\frac{1}{k} \ln|g + kv| + c_1 \\
 (t = 0, v = u) \\
 c_1 &= \frac{1}{k} \ln|g + ku| \Rightarrow t = -\frac{1}{k} \ln \left| \frac{g + kv}{g + ku} \right| \\
 \therefore \frac{g + kv}{g + ku} &= e^{-kt} \\
 \therefore g + kv &= (g + ku) e^{-kt} \Rightarrow v = \frac{1}{k} [(g + ku) e^{-kt} - g] \\
 x &= \int \frac{1}{k} [(g + ku) e^{-kt} - g] dt \\
 &= \frac{1}{k} \left[\frac{g + ku}{-k} e^{-kt} - gt \right] + c_2 \\
 (t = 0, x = 0) \\
 \therefore c_2 &= \frac{g + ku}{k^2} \\
 x &= \frac{1}{k} \left[-\frac{g + ku}{k} e^{-kt} - gt \right] + \frac{g + ku}{k^2} \\
 &= \frac{g + ku}{k^2} (1 - e^{-kt}) - \frac{gt}{k}
 \end{aligned}$$

QED

(ii) The two particles meet when $x_1 = x_2$

[NB You are allowed to assume the formula for x_2 !]

$$\text{ie } \frac{g+ku}{k^2}(1-e^{-kt}) - \frac{gt}{k} = h + \frac{g}{k^2}(1-e^{-kt}) - \frac{gt}{k}$$

$$\therefore \frac{g}{k^2}(1-e^{-kt}) + \frac{u}{k}(1-e^{-kt}) - \frac{gt}{k} = h + \frac{g}{k^2}(1-e^{-kt}) - \frac{gt}{k}$$

$$\therefore \frac{u}{k}(1-e^{-kt}) = h$$

$$\therefore 1-e^{-kt} = \frac{hk}{u} \Rightarrow e^{-kt} = 1 - \frac{hk}{u} = \frac{u-hk}{u}$$

$$\therefore -kt = \ln\left(\frac{u-hk}{u}\right) \Rightarrow kt = \ln\left(\frac{u}{u-hk}\right)$$

$$\therefore t = \frac{1}{k} \ln\left(\frac{u}{u-hk}\right)$$

- (b) (i) $ma = mv \frac{dv}{dx} = -(P + Qv^2) \Rightarrow a = v \frac{dv}{dx} = -\frac{1}{m}(P + Qv^2)$
- (ii) If $P = 0$ then $\frac{dv}{dx} = -\frac{Q}{m}v \Rightarrow \frac{dx}{dv} = -\frac{m}{Qv}$

If we transform the problem so that we take the *distance travelled* being from $x = 0$ (when $v = 3U/2$) to $x = D$ (when $v = U$) then

$$\int_0^D \frac{dx}{dv} dv = \int_0^D dx = -\frac{m}{Q} \int_{\frac{3U}{2}}^U \frac{dv}{v}$$

$$\therefore D = \left[-\frac{m}{Q} \ln|v| \right]_{\frac{3U}{2}}^U = -\frac{m}{Q} \ln \left(\frac{U}{\frac{3U}{2}} \right) = -\frac{m}{Q} \ln \left(\frac{2}{3} \right) = \frac{m}{Q} \ln \left(\frac{3}{2} \right)$$

QED

- (iii) If $P > 0$ then $\frac{dv}{dx} = -\left(\frac{P + Qv^2}{mv} \right) \Rightarrow \frac{dx}{dv} = -\frac{mv}{P + Qv^2}$

If we transform the problem so that we take the *distance travelled* being from $x = 0$ (when $v = U$) to $x = D$ (when $v = 0$) then

$$\int_0^D \frac{dx}{dv} dv = \int_0^D dx = -\int_U^0 \frac{mv dv}{P + Qv^2} = -\frac{m}{2Q} \int_U^0 \frac{2v dv}{P + Qv^2}$$

$$D = -\frac{m}{2Q} \left[\ln|P + Qv^2| \right]_U^0 = -\frac{m}{2Q} \ln \left(\frac{P}{P + QU^2} \right)$$

$$= \frac{m}{2Q} \ln \left(\frac{P + QU^2}{P} \right)$$

$$= \frac{m}{2Q} \ln \left(1 + \frac{Q}{P} U^2 \right)$$

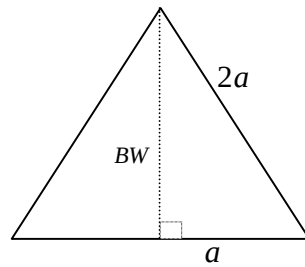
$$= \lambda \ln(1 + kU^2)$$

where $\lambda = \frac{m}{2Q}$ and $k = \frac{Q}{P}$

QED

Q7

(i)



By Pythagoras' Theorem

$$4a^2 = BW^2 + a^2$$

$$\therefore BW^2 = 3a^2$$

$$\therefore BW = \sqrt{3}a$$

(ii) Since $\triangle BWF \parallel \triangle VUF$

$$\therefore \frac{VU}{BW} = \frac{UF}{FW} \Rightarrow \frac{h}{\sqrt{3}a} = \frac{x}{b}$$

$$\therefore h = \frac{ax\sqrt{3}}{b}$$

(iii) Since $\triangle BWF \parallel \triangle VUF$ then $\frac{VF}{BF} = \frac{VU}{BW} = \frac{h}{\sqrt{3}a}$

$$BV = BF - VF$$

$$\triangle BLM \parallel \triangle RFS \text{ then } \frac{BV}{BF} = \frac{LM}{RS} \Rightarrow \frac{BF - VF}{BF} = \frac{LM}{a}$$

$$\therefore 1 - \frac{VF}{BF} = \frac{LM}{a} \Rightarrow 1 - \frac{h}{\sqrt{3}a} = \frac{LM}{a}$$

$$\therefore 1 - \frac{\frac{ax\sqrt{3}}{b}}{\sqrt{3}a} = \frac{LM}{a} \Rightarrow \frac{LM}{a} = 1 - \frac{x}{b} = \frac{b-x}{b}$$

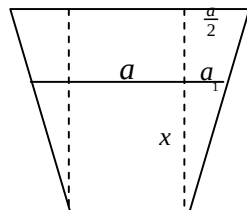
$$\therefore LM = \frac{a(b-x)}{b}$$

QED

(iv) **Clearly** when $x = 0$ then $PQ = a$ and when $x = b$ then $PQ = 2a$, so given the linear relationship of PQ in terms of x then

$$PQ - a = \frac{2a - a}{b}(x - 0) \Rightarrow PQ = \frac{a}{b}x + a$$

Alternative solution



$$PQ = a + 2a_1$$

$$\frac{a_1}{a/2} = \frac{x}{b} \Rightarrow 2a_1 = \frac{a}{b}x$$

$$\therefore PQ = \frac{a}{b}x + a$$

- (v) Area of slice is area of trapezium $KL MN$ and rectangle $KNQP$

$$KL MN = \frac{1}{2} \times \frac{ax\sqrt{3}}{b} \times \left[\frac{a(b-x)}{b} + \frac{a}{b}x + a \right]$$

$$= \frac{a^2 x \sqrt{3}}{b}$$

$$KNQP = a \times \left(\frac{a}{b}x + a \right) = a^2 \left(\frac{x}{b} + 1 \right)$$

So cross sectional area is given by

$$\frac{a^2 x \sqrt{3}}{b} + a^2 \left(\frac{x}{b} + 1 \right)$$

$$= \frac{a^2 x \sqrt{3}}{b} + a^2 \left(\frac{x+b}{b} \right)$$

$$= \frac{a^2 \left[x(1+\sqrt{3}) + b \right]}{b}$$

$$= \frac{a^2}{b} \left[x(1+\sqrt{3}) + b \right]$$

So the cross sectional volume is $\frac{a^2}{b} \left[x(1+\sqrt{3}) + b \right] \Delta x$

So the volume, V , is given by $\int_0^b \frac{a^2}{b} \left[x(1+\sqrt{3}) + b \right] dx$

$$V = \frac{a^2}{b} \int_0^b \left[x(1+\sqrt{3}) + b \right] dx$$

$$= \frac{a^2}{b} \left[(1+\sqrt{3}) \frac{x^2}{2} + bx \right]_0^b$$

$$= \frac{a^2}{b} \left[(1+\sqrt{3}) \frac{b^2}{2} + b^2 \right]$$

$$= \frac{a^2 b}{2} (3 + \sqrt{3})$$

[**NB** This is not a solid formed by rotation, so π *shouldn't* appear in the answer!]

Q8

(a)

Method 1	Method 2
$\frac{1}{a} + \frac{1}{b} - \frac{4}{t} = \frac{1}{a} + \frac{1}{b} - \frac{4}{a+b}$ $= \frac{b(a+b) + a(a+b) - 4ab}{ab(a+b)}$ $= \frac{a^2 - 2ab + b^2}{ab(a+b)}$ $= \frac{(a-b)^2}{ab(a+b)}$ ≥ 0 $\therefore \frac{1}{a} + \frac{1}{b} \geq \frac{4}{t}$	$(\sqrt{a} - \sqrt{b})^2 \geq 0 \Rightarrow a + b \geq 2\sqrt{ab}$ $\therefore \frac{1}{a+b} \leq \frac{1}{2\sqrt{ab}} \Rightarrow \frac{1}{\sqrt{ab}} \geq \frac{2}{a+b}$ Also $\frac{1}{a} + \frac{1}{b} \geq \frac{2}{\sqrt{ab}}$ $\text{So } \frac{1}{a} + \frac{1}{b} \geq \frac{2}{\sqrt{ab}} \geq \frac{4}{a+b} = \frac{4}{t}$
Method 3 (reductio ad absurdum)	

Assume $\frac{1}{a} + \frac{1}{b} < \frac{4}{t}$

$$\therefore \frac{a+b}{ab} < \frac{4}{t}$$

$$\therefore (a+b)^2 < 4ab \quad (\because t = a+b)$$

$$\therefore (a+b)^2 - 4ab = (a-b)^2 < 0$$

This last statement is clearly a contradiction as $k^2 \geq 0, k \in \mathbb{R}$

So the original assumption was false

$$\therefore \frac{1}{a} + \frac{1}{b} \geq \frac{4}{t}$$

- (b) (i) The total number of different outcomes:
 The first book can go in any of n boxes, so there is a total of n^n different arrangements.
 If there are to be no empty boxes, then the first book can go in any of n boxes, the next book only has $n-1$ boxes and so on. A total of $n!$
 So the probability of no empty box is $\frac{n!}{n^n}$
- (ii) For exactly one empty box, one box must have 2 books in it.
 So we have to pick the empty box, this can be done in n ways.
 Then we have to pick the box to have the two books, this can be in done in $n-1$ ways.
 Then we have $\binom{n}{2}$ ways of picking the two books that will go in the one box, leaving $(n-2)!$ ways of arranging the other books.
 A total of $n \times (n-1) \times \binom{n}{2} \times (n-2)! = \binom{n}{2} n!$
 So the probability is $\frac{\binom{n}{2} n!}{n^n}$ or $\frac{n(n-1)n!}{2n^n} = \frac{(n-1)n!}{2n^{n-1}}$
- (iii) With $n+1$ books to be distributed, this can be done in n^{n+1} ways because the first book has n boxes, the second book has n boxes and so on until the $(n+1)^{\text{st}}$ book.
 With no box to be empty, 1 box must have 2 books in it.
 We can choose this book in n ways. We can choose the 2 books in $\binom{n+1}{2}$ ways. The remaining books can be distributed in $(n-1)!$ ways.
 A total of $n \times \binom{n+1}{2} \times (n-1)! = \binom{n+1}{2} n!$ ways.
 So the probability is $\frac{n! \binom{n+1}{2}}{n^{n+1}}$ or $\frac{n(n+1)!}{2n^{n+1}} = \frac{(n+1)!}{2n^n}$

(iv) With $n + 2$ books to be distributed over n boxes this can be done in n^{n+2} ways.

If no box is to be empty there are two cases:

Case 1: 1 box has 3 books in it;

Case 2: 2 boxes have 2 books in it.

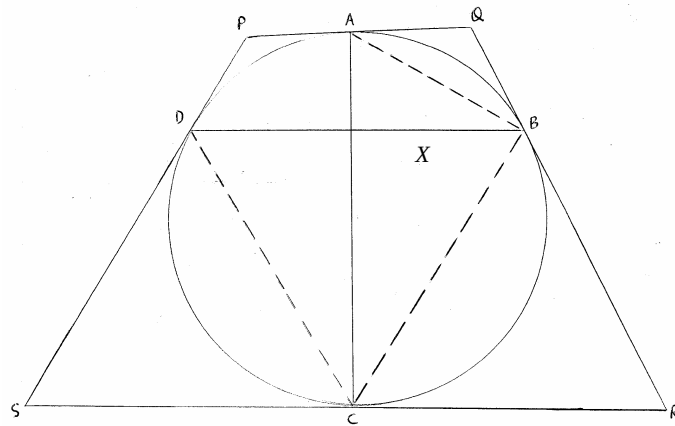
Case 1	Case 2
Pick the box to have 3 books, this can be done in n ways. Pick the 3 books, this can be done in $\binom{n+2}{3}$ ways. The remaining books can be distributed in $(n-1)!$ ways. A total of $\binom{n+2}{3} \times n \times (n-1)!$ ie $\frac{n(n+2)!}{6}$ ways	Pick the 2 boxes to have the 2 books this can be done in $\binom{n}{2}$ ways. Pick 2 books to go into the first of these boxes ie $\binom{n+2}{2}$ ways, then two books to go into the second box ie $\binom{n}{2}$ ways. Then the remaining books to be distributed in $(n-2)!$ ways. A total of $\binom{n}{2}^2 \times \binom{n+2}{2} \times (n-2)!$ ie $\frac{n(n-1)(n+2)!}{8}$ ways

So a total number of $\frac{(n+2)!}{6} + \frac{n(n-1)(n+2)!}{8}$ ways ie

$$\frac{4n(n+2)! + 3n(n-1)(n+2)!}{24} = \frac{n(3n+1)(n+2)!}{24} \text{ ways}$$

So the probability is $\frac{n(3n+1)(n+2)!}{24n^{n+2}} = \frac{(3n+1)(n+2)!}{24n^{n+1}}$

(c) (i)



NOT TO SCALE

Let $\angle S = 2x$, then $\angle Q = 180 - 2x$ ($PQRS$ is a cyclic quadrilateral)

Also $\triangle SDC$ is isosceles, so $\angle SCD = 90 - x$.

$\angle DBC = \angle SCD = 90 - x$ (alternate segment theorem)

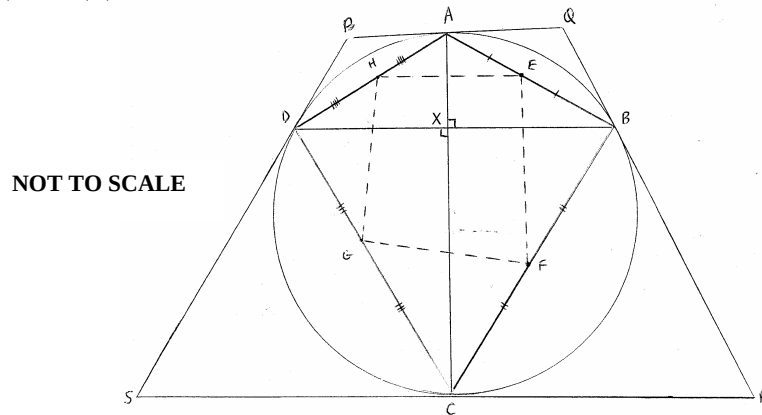
Similarly $\triangle SDC$ is isosceles, so $\angle QAB = x$.

Similarly $\angle BCA = \angle QAB = x$ (alternate segment theorem)

So $\angle CXB = 90^\circ$ (angle sum of triangle)

$\therefore AC \perp BD$ **QED**

(c) (ii)



NOT TO SCALE

Lemma: The midpoints of a quadrilateral form a parallelogram

Proof: $AH : HD = AE : EB = 1 : 1$

$HE \parallel DB$ (Midpoint Theorem for Triangles)

Similarly $GF \parallel DB \Rightarrow HE \parallel FG$

Similarly $HG \parallel AC$ & $AC \parallel EF \Rightarrow HG \parallel EF$.

$\therefore EFGH$ is a parallelogram. **QED**

$\therefore AC \perp BD$, $HE \parallel DB$ & $GF \parallel DB$ and $HG \parallel AC$ & $AC \parallel EF$

$\therefore \angle HGF = \angle GFE = \angle FEH = \angle EHG = 90^\circ$

$\therefore E, F, G$ and H are concyclic (All rectangles are concyclic)

QED